ITA0303 - mobile Computing

CO4 At2 - Case study

Name :- S. Hemaja

Regno :- 192373035

Android v/s iOS for a Security - Sensitive mobile Application.

Security Aspect	Android	iOS
Security Architecture	Based on the linux Kernel with multiple Security layers, including SELinux, application Sandboxing, permissions Verified boot, and Encryption.	use a tightly integrated hardware-software Security architecture with Secure boot, code signing, Sandboxing, data protection and hardware Security.
Application isolation	Each application normally runs under its own Linux UID and Sandbox. Android permissions Control access to Sensitive resources Such as contacts, camera, location and storage.	Each application runs in a strong Sand box and has restricted access to other applications and System resources. App store code-signing.

Runteme Envronment	use Anchoid runtime to execute applications ART provides memory management, runtime Security checks, and application isolation	uses Apple's runteme Environment. with strict code-signing, Sandboxing, memory protections, and controlled APIS.
System level protection	selinux enforces mandatory acess control. Verified boot helps prevente unauthorized modification of the OS. Hardware backed keystore protects cryptographic keys.	Secure Enclave, hardware-backed key protection, Secure boot, data protection Key chain, and code Signing provide strong System level Security.
Data protection	File based Encryption and Android keystore can protect Sensitive application data and Cryptographic keys.	Data protection automatically encrypts protected files, while keychain Securely stores credentials Keys and authentication information.

Application Distribution	Supports google play and depending on device configuration, installation from other source. This provides flexibility but can increase security risks.	App distribution is more tightly controlled through Apple's signing and review mechanisms, reducing exposure to unauthorized applications
Enterprise Security	Android Enterprise supports managed devices, work profiles, application policies, and Enterprise mobility management.	Provides strong Enterprise management through mobile device management managed applications, configuration policies, and device controls.
Main Advantage	Greater flexibility, customization, and wide hardware availability.	strong integrated security application isolation, hardware backed protection and controlled Ecosystem.
main Concern	Security can vary between manufactures, Android Versions, and device configurations.	More restrictive Ecosystem and less customization compared with Android.

Analyses.

1. Security Architecture:-

Android uses a layered security model build around the linux kernel, SE linux, Sand boxing, permissions Encryption.

2. Application Isolation:-

Both operating systems isolate Applications from each other. Android Assigns applications Separate linux identities.

3. Runtime Environment:-

Android Applications Execute using ART which provides runtime management and Security mechanisms.

4. System level protection:-

Android provides SE Linux, Verified boot, Android keystore, encryption, and hardware backed Security. iOS provides Secure Enclave, Secure boot, key chain, Data protection.